

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected December 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C/C++, Python, Java

Controls & Dynamics: Fixed-point PI/PID, saturation, anti-windup, slew limiting, state estimation, sensor fusion, Bode/Nyquist

Robotics & Embedded: PX4, visual-inertial odometry, IMU fusion, STM32H7, CAN, real-time firmware

Simulation & Validation: Software-in-the-loop, closed-loop simulation, scripted replay, regression tests, parameter sweeps, NumPy, pandas, Matplotlib

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead **12 engineers** across **10+ vehicle-control, modeling, validation, and telemetry projects** for a Formula SAE electric racecar
- Developing the custom-inverter VCU path for a **four-motor** EV, mapping calibrated pedal input to bounded CAN current commands on a **10 ms** loop with fail-safe precharge and ready-to-drive gating
- Lead battery, BMS, motor, and inverter plant-model integration for a core team of **7 engineers**, defining interfaces so VCU control code can be evaluated before hardware tests

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Tuned PX4 PID flight-control loops to reduce vertical and horizontal drift during autonomous position hold
- Integrated and tuned visual-inertial odometry for onboard state estimation, connecting perception output to the flight-control stack and evaluating changes against logged telemetry
- Developed embedded flight-control changes in C and iterated gains through repeatable hardware tests and log review

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Developed fixed-point PI firmware with saturation, anti-windup, and slew limiting that held measured power within **0.01%** of its commanded setpoint while enforcing an **80 kW** limit
- Built Python telemetry parsers and a software-in-the-loop replay harness for recorded CSV logs, controller parameter sweeps, and regression testing
- Planned and executed on-vehicle tests, measured control response and latency, and generated plots for tuning and baseline comparison

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Tuned the PI torque controller to within **2%** error using logged step responses, and the vehicle placed **1st in endurance** at MIS EV 2025

PROJECT EXPERIENCE

SR-Wireless-CAN, Control-System Test Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, CAN, recorded-trace replay, FastAPI, WebSockets, SQLite, Docker

- Led backend development for a test platform supporting **3 capture modes**, including simulated data, recorded-trace replay, and live CAN, plus **3 signal-monitoring views**
- Reverse-engineered and implemented an **8-stage** CAN firmware-deployment workflow from vendor traces, with acknowledgment checks, CRC verification, and persisted run history